

Set Theory & Functions: Study Notes

Topics: partitions & equivalence relations · partial order relations (posets) · bounds, chains & well-ordering · functions & composition

1. Partitions

In plain words: a partition splits a set into separate groups so that every item is in exactly one group — no item left out, no item in two groups. Like sorting a deck of cards by suit: every card goes in exactly one pile, and all the piles together make up the whole deck.

DEFINITION

Let A be a set and $\mathcal{A} = \{A_n : n \in I\}$ a collection of non-empty subsets $A_n \subseteq A$. We say A is *partitioned by $\mathcal{A}$* if:

1. $A = \bigcup_{n \in I} A_n$ — the A_n together make up A .
2. If $A_m \neq A_n$, then $A_m \cap A_n = \emptyset$ — distinct A_n are *pairwise disjoint*.

Equivalently, $\mathcal{A}$ is a partition of A iff: $\mathcal{A} \subseteq \mathcal{P}(A)$, $\bigcup \mathcal{A} = A$, and $\mathcal{A}$ is pairwise disjoint.

EXAMPLE

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$ with $A_0 = \{1, 2, 5\}$, $A_1 = \{3\}$, $A_2 = \{4, 6, 7\}$. Then $\mathcal{A} = \{A_0, A_1, A_2\} \subseteq \mathcal{P}(A)$, $A = A_0 \cup A_1 \cup A_2$, and $\mathcal{A}$ is pairwise disjoint — so $\mathcal{A}$ is a partition of A .

THEOREM — PARTITIONS $\leftrightarrow$ EQUIVALENCE RELATIONS

Let A be a set.

1. If $\sim$ is an equivalence relation on A , then A is partitioned by the equivalence classes of $\sim$.
2. If $\{A_n\}_{n \in I}$ is a partition of A , then the relation $\sim$ on A defined by $a \sim b \iff \exists n \in I$ such that $a \in A_n$ and $b \in A_n$ is an equivalence relation.

These two constructions are inverse to each other — every partition comes from exactly one equivalence relation, and vice versa.

2. Partial Order Relations (Posets)

In plain words: a partial order is a way to say "this comes before that" that doesn't force every pair of things to be comparable. Three simple rules make it work: everything relates to itself, if two different things "point" both ways they must actually be equal, and the order carries through a chain (if a is before b , and b is before c , then a is before c). "Partial" just means some pairs are allowed to be unrelated — neither one comes before the other.

DEFINITION

A relation $\mathcal{R}$ (often written $\leq$) on A is a *partial order relation* if it is simultaneously *reflexive*, *antisymmetric*, and *transitive*. A set A on which a partial order is defined is called a *poset*, written $(A, \leq)$.

For all $a, b \in A$, the notation $a < b$ means $a \leq b$ and $a \neq b$ (strict order).

Examples: $\leq$ is a partial order on $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{R}$. The relation $\mathcal{R} = \{(a,b) \mid a \subseteq b\}$ on $\mathcal{P}(A)$ for $A = \{1,2,3\}$ is a partial order. Both $\leq$ and $=$ are partial orders on $\mathbb{R}$, but $<$ is *not* a partial order on $\mathbb{R}$ (it fails reflexivity). An equivalence relation is generally not a partial order.

Remark (comparability). "Partial" indicates that not every pair of elements need be comparable: $a, b \in A$ are *comparable* if $a \mathcal{R} b$ or $b \mathcal{R} a$. E.g. in $(\mathcal{P}(\{1,2,3\}), \subseteq)$: $\{1,3\}$ and $\{2\}$ are *not* comparable (neither is a subset of the other), while $\{1\} \subseteq \{1,3\}$ are comparable.

THEOREM — WEAK TRICHOTOMY LAW

If $\leq$ is a partial order on A , then for all $a, b \in A$, *at most one* of the following holds: $a < b$, $b < a$, or $a = b$.

Proof. Suppose $a < b$ (so $a \leq b$ and $a \neq b$). If also $b < a$, transitivity gives $a < a$ — a contradiction, so $b < a$ and $a = b$ are both precluded. Symmetric reasoning rules out $b < a$ jointly holding with the others. If $a = b$, then by definition of $<$ neither $a < b$ nor $b < a$ can hold. ■

3. Bounds, Minimal & Maximal Elements

In plain words: the *least* element must be below absolutely everything — a strict requirement, so at most one element can have this title. A *minimal* element is more relaxed — it just means nothing sits below it, but there can still be other things it's simply not compared to. So a poset can have several minimal elements but never more than one least element.

DEFINITION — LEAST / GREATEST ELEMENT

Let $(A, \leq)$ be a poset and $m \in A$.

- m is the *least element* of A if $m \leq a$ for *all* $a \in A$.
- m is the *greatest element* of A if $a \leq m$ for *all* $a \in A$.

Least/greatest elements, when they exist, are unique — if m and m' both satisfy the least-element condition, then $m \leq m'$ and $m' \leq m$, so antisymmetry forces $m = m'$ (same argument for greatest). This uniqueness is what justifies calling it *the* least/greatest element. There is no guarantee a poset has either.

EXAMPLE

A finite set of reals has both a least and greatest element under $\leq$. $\mathbb{Z}^+$ has a least element (1) but no greatest. Both $\{5n : n \in \mathbb{Z}^-\}$ and $\mathbb{Z}^-$ have a greatest element but no least. The set $B = \{5n : n \in \mathbb{Z}^+\}$ has least element 5, since $5 \in B$ and $5(1) \leq 5n$ for all $n \in \mathbb{Z}^+$.

DEFINITION — MINIMAL / MAXIMAL ELEMENT

Let $(A, \leq)$ be a poset and $m \in A$.

- m is a *minimal element* if $a \not\leq m$ for all $a \in A$ (nothing is strictly below m).
- m is a *maximal element* if $m \not\leq a$ for all $a \in A$ (nothing is strictly above m).

Unlike least/greatest, minimal and maximal elements need not be unique, since not all elements need be comparable.

DEFINITION — COMPARABLE / INCOMPARABLE

Let $(A, \leq)$ be a poset, $a, b \in A$. If $a \leq b$ or $b \leq a$, then a and b are *comparable*. Elements not comparable are *incomparable*.

4. Chains, Linear Orders & Compatibility

In plain words: a chain is a group where every pair can be compared — like a straight line-up, you can always say which one comes first. A linear order is when the *whole* poset is one big chain (like $\mathbb{R}$ under $\leq$). Compatibility is a looser idea: two elements don't have to be comparable, they just need something below both of them. For example, 4 and 6 aren't directly comparable by division, but they're still compatible because 1 and 2 both divide them.

DEFINITION — CHAIN

A subset C of poset $(A, \leq)$ is a *chain* if a is comparable to b for *all* $a, b \in C$ — i.e., every pair of elements inside a chain is comparable.

EXAMPLE

Let $A_k = \{x \in \mathbb{Z} : (x-1)(x-2)\cdots(x-k) = 0\}$ for $k \in \mathbb{Z}^+$. Take $m, n \in \mathbb{Z}^+$: $A_m = \{1, 2, \dots, m\}$ and $A_n = \{1, 2, \dots, n\}$. If $m \leq n$ then $A_m \subseteq A_n$, otherwise $A_n \subseteq A_m$ — so every pair is comparable and $\{A_k : k \in \mathbb{Z}^+\}$ is a chain with respect to $\subseteq$.

DEFINITION — LINEARLY ORDERED SET

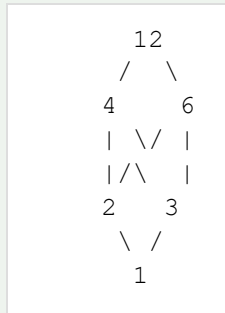
A poset $(A, \leq)$ is a *linearly ordered set* and $\leq$ is a *linear order* if A itself is a chain with respect to $\leq$ (i.e. every pair of elements is comparable). Since every subset of $\mathbb{R}$ is a chain under $\leq$, $\leq$ is a linear order on $\mathbb{R}$.

DEFINITION — COMPATIBLE / INCOMPATIBLE

Let $(A, \leq)$ be a poset. $a, b \in A$ are *compatible* if there exists $c \in A$ such that $c \leq a$ and $c \leq b$ (a common lower bound). If not, a and b are *incompatible*, written $a \perp b$.

EXAMPLE 1 — DIVISIBILITY POSET

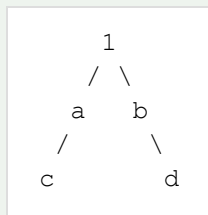
Let $A = \{1, 2, 3, 4, 6, 12\}$ ordered by divisibility: $a \leq b \Leftrightarrow a \mid b$.



Consider $a = 4$, $b = 6$. Since $1 \leq 4$ and $1 \leq 6$, the element 1 is a common lower bound (also $2 \leq 4$, but $2 \nmid 6$). Hence 4 and 6 are **compatible**.

EXAMPLE 2 — A SLIGHTLY LARGER POSET

Consider the poset below, order upward, with $c \leq a$, $d \leq b$:



The elements a and b have *no* common lower bound (c is below a only, d is below b only, and 1 is above both, not below). Therefore a and b are **incompatible**.

5. Well-Ordered Sets

In plain words: a well-ordered set is a linear order where you can always pick out a smallest member, no matter which subset you look at. That's true for $\mathbb{N}$ (whole numbers 1, 2, 3, ...) but not for $\mathbb{R}$ — the set of positive real numbers has no smallest member, because you can always find a smaller one.

DEFINITION

A linearly ordered set $(A, \leq)$ is a *well-ordered set* and $\leq$ is a *well-order* if every nonempty subset of A has a least element with respect to $\leq$.

$(\mathbb{N}, \leq)$ is a well-ordered set (this fact about the natural numbers is taken as an axiom, not proved) — giving us at least one well-ordered set to work with.

THEOREM

If $(A, \leq)$ is well-ordered and B is a nonempty subset of A , then $(B, \leq)$ is well-ordered.

Proof. Let $B \subseteq A$, $B \neq \emptyset$. To show B is well-ordered, let $C \subseteq B$, $C \neq \emptyset$. Then $C \subseteq A$. Since $\leq$ well-orders A , C has a least element with respect to $\leq$. ■

6. Functions

In plain words: a function is just a rule that takes every input and gives it exactly one output — no input skipped, no input sent to two different places. The codomain is the "target space" you announce ahead of time (like "all real numbers"); the range is only the part actually reached by some input, so range is always inside the codomain, sometimes a smaller part of it.

DEFINITION

A *function* is an ordered triple (f, A, B) such that:

1. A and B are sets, and $f \subseteq A \times B$,
2. for every $x \in A$ there is some $y \in B$ such that $(x, y) \in f$,
3. if $(x, y) \in f$ and $(x, z) \in f$, then $y = z$ (the assignment is *unique*).

A is the *domain* of f , B its *codomain*. We write $f : A \rightarrow B$. If $(x, y) \in f$ we write $y = f(x)$, and call y the *image* of x under f . The *range* of f is $\{y \in B : \exists x \in A, y = f(x)\}$ — note $\text{range}(f) \subseteq B$, but they need not be equal.

Notation. If $U \subseteq A$, the image of U is $f(U) = \{f(u) \in B : u \in U\}$. The image of all of A is precisely the range: $f(A) = \text{range}(f) = \text{Im}(f) = \{f(a) : a \in A\}$.

EXAMPLE 1 — VALID FUNCTION

$f = \{(1,a), (2,b), (3,a)\}$ from $A = \{1,2,3\}$ to $B = \{a,b,c\}$. Since $D(f) = A$ and every element of A has a unique image, f is a function. Range: $R(f) = \{f(1), f(2), f(3)\} = \{a, b\}$. Note $R(f) \subseteq B$.

EXAMPLE 2 — FAILS UNIQUENESS

$f = \{(1,a), (2,b), (2,c), (3,a)\}$ is *not* a function: $2 \in A$ has two images (b and c), violating uniqueness.

EXAMPLE 3 — FAILS "FOR EVERY"

$f = \{(2,b), (3,a)\}$ on $A = \{1,2,3\}$ is *not* a function: $D(f) = \{2,3\} \neq A$, so element 1 has no assigned image.

EXAMPLE 4

$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ is a function. $D(f) = \mathbb{R}$, and $R(f) = \{x^2 \mid x \in \mathbb{R}\} = \{x \mid x \in \mathbb{R} \text{ and } x \geq 0\}$.

Graph of a Function & the Vertical Line Test

The *graph* of f is the set of ordered pairs $\{(x, f(x)) \mid x \in D(f)\}$. *Vertical line test*: a curve in the xy -plane is the graph of a function iff every vertical line intersects it at most once (otherwise uniqueness is violated for some x).

Equality & Composition of Functions

In plain words: composition just means feeding the output of one function into another, like a pipeline. It only works if what f puts out is something g knows how to accept. When you chain three or more functions, it never matters how you group them — only the order matters.

DEFINITION — EQUAL FUNCTIONS

$f : A \rightarrow B$ and $g : C \rightarrow D$ are *equal* iff (1) $A = C$, (2) $B = D$, and (3) $f = g$, meaning $f(x) = g(x)$ for every $x \in A$.

DEFINITION — COMPOSITION

Let $f : A \rightarrow B$ and $g : C \rightarrow D$ with $\text{ran } f \subseteq \text{dom } g$. The *composition* $g \circ f : A \rightarrow D$ is defined by $(g \circ f)(x) = g(f(x))$.

Domain: $D(g \circ f) = \{x \mid x \in D(f) \text{ and } f(x) \in D(g)\}$.

LEMMA 1

If f and g are functions with $\text{ran } f \subseteq \text{dom } g$, then $\text{dom}(g \circ f) = \text{dom } f$, and $\text{codom}(g \circ f) = \text{codom } g$.

LEMMA 2 — ASSOCIATIVITY

Let $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$ be functions. Then $h \circ (g \circ f) = (h \circ g) \circ f$ — composition of functions is associative. (Proof sketch: set $p = h \circ (g \circ f)$ and $q = (h \circ g) \circ f$, then show p and q are equal as functions by matching domain, codomain, and pointwise values.)

Quick-Reference Summary

Concept	Key Condition
Partition of A	$\mathcal{A} \subseteq \mathcal{P}(A)$, $\bigcup \mathcal{A} = A$, pairwise disjoint
Partial order (poset)	reflexive + antisymmetric + transitive
Least / greatest element	below / above every element — unique if it exists
Minimal / maximal element	nothing strictly below / above it — not necessarily unique
Chain	every pair of elements is comparable
Linear order	the whole poset is a chain
Well-order	linear order + every nonempty subset has a least element
Compatible a, b	share a common lower bound c
Function $f : A \rightarrow B$	$f \subseteq A \times B$, total (every $x \in A$ assigned) & single-valued (unique y)
$g \circ f$	$(g \circ f)(x) = g(f(x))$, requires $\text{ran } f \subseteq \text{dom } g$; associative